Workplace Handbook for SciLifeLab Campus Solna

November 2025, Version 6.1

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Updated sections from previous version June 2024

- *References to the Swedish Work Environment authority's provisions have been updated.*
- *2.7 Safety Round and Work Environment Evaluation*
- *8.3 Fire safety organisation – reformulated the evacuation leader role*
- *8.4 Fire/evacuation alarm*
- *8.5 Floor evacuation alarm – removed*
- *9.4 Chemical waste – Empty Gas cartridges added*
- *Appendix 1: Introduction checklist*

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1. Introduction

This handbook is essential for all personnel, whether in laboratory or non-laboratory roles, at SciLifeLab Campus Solna. Its primary goal is to enhance awareness of workplace risks and provide comprehensive guidelines for safety and emergency procedures. We strongly encourage all personnel to review this document annually to ensure a thorough understanding of safety measures.

SciLifeLab is a national infrastructure and platform for collaboration for molecular and computational life science research across Swedish universities. At SciLifeLab Campus Solna the collaborating universities are Royal Institute of Technology (KTH), Karolinska Institute (KI), Stockholm University (SU), and Uppsala University (UU).
SciLifeLab is not an employer, and most operational regulations are dictated by your respective host university. This includes finance, human resources, procurement, travel management, insurance, and more.

It is your home university that bears the responsibility for the health and safety of their personnel stationed at SciLifeLab CS. Recognizing the importance of ensuring and coordinating a safe work environment for all personnel at SciLifeLab CS, the three host universities - KI, KTH and SU- have signed a cross-university agreement regarding work environment and fire safety (Dnr: HS-2025-0556).
This agreement serves as a framework for effectively regulating and upholding the standards and responsibilities of workplace safety. The Site Support team at SciLifeLab CS (SiSu) is responsible for coordinating this work and must therefore be informed of any changes that can affect the work environment.

This handbook has been developed in collaboration with work environment representatives from all host universities and provides vital information on shared routines and safety regulations. All colleagues based at Campus Solna are expected to adhere to these shared procedures.

In addition to the common guidelines outlined in this handbook, please be aware that there are additional rules and regulations specific to your home university. Adherence to your home university's policies and keep yourself informed of any updates or changes through your home university's communication channels.

It you have any queries or require assistance during your tenure, do not hesitate to reach out to the Site Support team. A warm welcome to SciLifeLab Campus Solna!

/ Your Site Support Team

2. Administrative

2.1 Site Support

The Site Support (SiSu) at SciLifeLab Campus Solna administers the local facilities and guides and coordinates all the activities within SciLifeLab Campus Solna, together with other important functions such as the IT department and the facility services provided, such as housekeeping, deliveries and the reception. You can read more about the SiSu team and what we do on the intranet: http://intranet.scilifelab.se/contact/contact-us

Location: Gamma Floor 7, office G7160

You can contact Site Support via our ticket system: https://intranet.scilifelab.se/support/ or at support@scilifelab.se.

2.2 Introduction New Personnel

All new personnel should be properly introduced to the SciLifeLab CS premises and to the lab safety routines prior to starting their work. All newcomers must have an appointed supervisor who is responsible for making sure that the newcomer gets the proper introductions, as well as supervising and helping the new members to get started with their activities. The supervisor informs about the matters as defined in the SciLifeLab introduction checklist (*See Appendix 1. Introduction checklist)* within the first week. When the checklist has been completed, it should be signed by both the introducer and the newcomer and be kept in the lab for reference.

It is strongly recommended that all new members read the SciLifeLab Workplace Handbook prior to starting their work.

Please note that this handbook does not cover all the safety routines, please make sure that the new lab members are also introduced to;

- The general risks in the workplace as well as specific risks.
- How risk assessments are used and where to find them.
- Information on any specific waste handling.
- Relevant equipment, methods and procedures.

SciLifeLab Campus Solna provides a safety introduction tailored for all personnel, whether in laboratory or non-laboratory roles, specific to our campus. All individuals stationed at SciLifeLab CS are expected to participate in this safety introduction as soon as it becomes available. Additionally, depending on your affiliation, you may be required to undergo safety education from your home university. For further details regarding safety introductions, please contact Site Support or the administration within your respective school or department.

2.3 Responsible for Workplace and Laboratory Safety

SciLifeLab has no organizational number and is thus not a legal entity. All personnel are employed by one of the Stockholm trio universities. The employer has the main responsibility for the workplace and laboratory safety. Make sure you know the division of responsibilities at your home university.

- Group Leader

The group leader, or their equivalent, holds official staff liability, which also encompasses responsibility for students. Other terms used include immediate manager (*Swedish: närmaste chef*). It is the responsibility of the group leader to:

- provide safe working conditions for all personnel and ensure that safe work practices are followed.
- continually educate all personnel on the potential hazards associated with specific laboratory practices and of the precautionary measures relevant to the hazards (e.g. personal protective equipment).
- forward important information sent out by Site Support to all group members.
- meet the legal requirements of governmental legislation for occupational health and safety, and implement new work practices or policies recommended by the Swedish Work Environment Authority, and other relevant authorities.
- investigate accidents and initiate corrective actions, which ensure safe working conditions.

- The Employees and Students

Individual employees, laboratory workers and students are obligated to;

- following Good Laboratory Practices *(see section 3.4 Good Laboratory Practices).*
- use appropriated Personal Protective Equipment (PPE) such as lab coat, gloves, safety glasses… in the laboratory.
- planning and conducting each operation in accordance with the operating procedures outlined in this workplace Handbook (e.g., complete risk assessments for your work before starting any experiment).
- report unsafe acts or conditions to the principal investigator, Safety representative or to the SciLifeLab Lab Safety Coordinator.
- familiarize themselves with evacuation routes, locations of fire extinguisher equipment, eye wash stations and alarm buttons. This also includes knowing how these work and how to behave in case of an incident.

- Safety representatives (Skyddsombud)

The Safety representatives at SciLifeLab represent the employees in health and safety issues. The Safety representatives work for a risk-free work environment and their main aim is to prevent accidents, injuries, and work-related illnesses. It is important to know that the Safety

Representative has no responsibility for the work environment and can never be held accountable for not taking actions. That responsibility falls back on the employer (the group leader).

Safety representatives monitor the laboratory activities and the work areas to ensure that (1); workers comply with the SciLifeLab policies and governmental safety regulations and that (2); there are no safety hazards present in the work area.

Each floor at SciLifeLab has at least one appointed Safety representative. In case of any workplace accident or incident, or any other work environment related issue or concern, please contact your closest Safety representative. In case of an incident or accident, the Safety representative will help you to make a report of the event, and also help to investigate the cause and recommend actions such as changes in routines, in order to prevent future incidents.

- Floor Representatives

Each floor has a designated floor representative. Their task is to monitor the general status of the floor and organize floor meetings, which is a forum for the floor to appoint responsibilities within the unit (for example Safety representatives and evacuation leaders) and to organize floor cleaning days and kitchen lists. The floor meetings are an opportunity for the floor staff to discuss work environment related issues or questions.

Moreover ,there are people on every floor in charge of different task related to fire safety. They are a part of the fire safety organization at SciLifeLab CS

More information on these different roles and contact details can be found on the SciLifeLab intranet: http://intranet.scilifelab.se/working-at-scilifelab/for-your-work/responsibility-areas/ and in the section *8.2 Fires safety organization.*

2.4 Occupational Health Care Services

All universities provide occupational health care services for their employees. All employees are entitled to telephone counselling regarding work related injuries or questions/concerns. You can talk to mental health-, legal-, or financial experts. The health care service often also provides services such as;

- One or two visits per year to a nurse, physiotherapist, or other health specialist.
- Work related vaccinations (e.g., Hepatitis).
- Treatment of puncture wounds and blood tests.
- Health assessments for work-related exposures.

Students may receive support from the university’s Student Health Service.

You can read more about your university’s health care services on your home university staff pages.

2.5 Reporting incidents, risks and occupational injuries

All workplace incidents, risks and occupational injuries should be reported. This includes if you have been involved in or witnessed some kind of unsafe situation. It is important that all incidents/risks and occupational injuries that occur at the workplace are handled. The better we become at investigating and addressing the causes, the better we will become at preventing such incidents.

Examples of incidents, risks and occupational injuries are: fraud, harassment, wrongly labelled hazardous waste, cut injuries, slipping/falling, chemical spill, exposure to chemicals etc.

Incidents, risks and occupational injuries are reported via the incident reporting systems at your home university. Ask your closest supervisor and Safety representative for help to make an official incident report.

According to Swedish Law, serious work injuries and incidents must be reported by the employer to the Work Environment Authority (Arbetsmiljöverket) within 48 h of the incident. Examples of serious incidents are bone fractures, extensive bleeding, unconsciousness/concussion, serious threat to an employee/student, known or strong suspicion of infection with cuts/puncture wounds or burns (2nd-3rd degree).

2.6 Courses in Work Environment

SciLifeLab Campus Solna and your home university offer courses in work environment and safety. All new personnel must attend the SciLifeLab Campus Solna Safety Introduction as soon as one is offered. All personnel must have attended a Fire Safety introduction no longer than 4 years ago. Every group must have enough personnel trained in cardiopulmonary resuscitation (CPR) at the workplace.

2.7 Safety Round and Work Environment Evaluation

Once a year, each group or unit must conduct an annual work environment evaluation and a physical safety round. It is the responsibility of the group leader to ensure that these are carried out. Groups that collaborate are encouraged to carry out their evaluations and inspections jointly.

Campus Solna Site Support will send a yearly reminder to all group leaders including instructions and documentation templates for the evaluation and safety round. Site Support is also responsible for conducting regular safety rounds of common areas together with the university's main safety representatives.

Every third year, Campus Solna Site Support organizes and participates in safety rounds with all groups and units at Campus Solna. Site Support will schedule, make invitations and lead the safety rounds covering all floors and spaces on Campus Solna.

For more information, please visit Safety Round and Work enviroment Evaluation on the intranet.

3. General Information

3.1 Pregnant and Breastfeeding Women

There are special rules for pregnant and breastfeeding women and a separate risk assessment is necessary to assess the work duties. As soon as the employer is informed, he/she must offer the risk assessment to the affected personnel/student. A specific risk assessment form is available on the intranet and can be used to assess ergonomics, psychosocial, chemical, biological, and physical risks. Discuss your work with your group leader or a Safety representative. Please note that during pregnancy the main formation of organs is during the first 8 weeks. The first 3 months are therefore considered as the most sensitive pregnancy period of external influences.

3.2 Resting rooms

There are three resting rooms (vilrum) at Campus Solna. They are located in the basement (A1751, key available at the goods reception), Gamma 6 (G6350) and Gamma 7 (G7351). They may be used if you feel unwell or need to rest during the workday. For safety reasons, the person using the resting room should ALWAYS inform a colleague and write down the check-in time on the whiteboard on the door. In case of sickness, the coworkers should check up on the person in the resting room regularly. We recommend that you use the resting room for no more than one hour. If you are still not feeling well, you should go home/seek assistance. Never use the resting room when working alone.

3.3 Working alone

Solitary work refers to tasks performed by workers in physical and/or social isolation from other people. Physical isolation means that the worker cannot contact others without using technical communication devices. Social isolation means that the employee cannot rely on help from others in critical or risky situations due to the working conditions (AFS 2023:2, 6 chap. on solitary work).

Solitary work should be limited and, if possible, completely avoided.

The group leader in consultation with the Safety representative and affected person determines which work is to be considered so risky that it must not be carried out alone. If working alone cannot be avoided, the group leader is responsible to carry out a risk assessment in consultation with the Safety representative and the involved coworker. The risk assessment shall provide concrete measures that need to be implemented to ensure that working alone can be carried out safely. Rules for hazardous work, especially if it is performed alone, must be documented and clarified to all employees.

Exceptions:

- Working alone with liquid nitrogen is forbidden.
- Students are not allowed to work alone in the lab outside of normal office hours (8.00-17.00).

3.4 Rules for Minors

Special rules apply to minors in practice or visiting SciLifeLab Campus Solna activities. For instance, minors are not allowed to work with experimental cancer research or certain hazardous substances/agents (see Appendix 2 in AFS 2023:2). Find more information on the intranet: https://intranet.scilifelab.se/prao-students-and-minors-at-scilifelab/

3.5 Good Microbiological Practices and Procedures

Some general rules for our common laboratory areas;

- All shared laboratory areas (such as the chemical storage rooms and the sculleries) have to be cleaned immediately after usage and all things brought there must be removed when done.
- No laboratory gloves or lab coats should be worn outside of the laboratories (gloves and coats are not allowed in the staircases, elevators, toilets, lunchrooms or office spaces).
- As a general practice: never touch door handles while using laboratory gloves!
- Lab reagents, waste bins and other lab related materials should always be transported in the goods elevator.

Good Laboratory Practices (GLPs) are to be followed in the laboratory areas at all times;

- Do not eat, drink, apply cosmetics, use tobacco products, or handle food in the lab.
- It is obligatory to wear protective clothing (lab coat) in the lab, but never outside in the office areas, staircases, or elevators, etc.
- Avoid using noise-cancelling headphones in the lab, as they can prevent you from hearing important alarms, warnings, or communications.
- Observe cleanliness and maintain good order in the lab.
- Avoid spills, splashes and formation of aerosols when handling liquids or powders.
- Handle needles and sharps safely. Never recap them and dispose of these in approved sharps containers.

- Manage laboratory waste safely and responsibly. Use correct bins and labelling.
- Do not put yourself or your colleagues in danger.
- If you notice any misconduct, unsafe behaviors, or work environment hazard, inform your closest Safety representative or supervisor.
- Pets are not allowed at SciLifeLab.
- Remember: it is OK to make mistakes! If you are unsure of how to safely handle a situation, material, or of the risk associated with any procedure, always seek guidance!

3.6 Autoclaves, MilliQ and Dishwasher Stations

Common autoclave- and dishwasher stations are located at Alfa 2, Alfa 4, Gamma 3, Gamma 4 and Gamma 5. In most of these locations you can also find ice machines and MilliQ water (ultrapure water).

All laboratories are equipped with deionized high purity water (Green taps) which can be used for most protocols. The high purity water in the labs is filtered and treated the same way as the MilliQ-water and should therefore be of the same quality.

You must be introduced to the use of the common equipment by your supervisor or an experienced colleague before using it for the first time. In case of any problems with the common equipment in the autoclave- and dishwasher stations, please contact Site Support.

3.7 Common -80°C /-140°C Freezer Rooms

All groups at SciLifeLab have the possibility to rent freezer space in the common -80°C and -140°C freezers, and to rent space to store extra -20°C freezers and +4°C fridges.

The common SciLifeLab freezers are located mainly in these rooms;

- Room **A1691**, Alpha building, first floor.
- Room **A6200**, Alpha building, sixth floor.
- Room **A7800**, Alpha building, seventh floor.
- Room for -20°C freezers: **A7820**, Alpha building, seventh floor.

In order to get access to the common freezer rooms you need to read the freezer routines and finish a freezer quiz. Most administration regarding the common freezer rooms are handled by the Freezer Core Team, which consists of volunteer members from the laboratories at SciLifeLab.

More information on the Freezer Core Team and how to get access:

intranet.scilifelab.se/lab-orders/common-freezers-and-freezer-rooms

For questions regarding the common freezer rooms: freezercoreteam@scilifelab.se

If a group needs more space than provided by SciLifeLab, the group has to purchase the freezer themselves, and the freezer has to be located in their private lab space.

3.8 Supply Centers

There are two self-service shops at SciLifeLab:

- Room G1500, Gamma floor 1 – A supply center for consumables, kits, media and reagents.
- Room A1773, Alfa floor 1 – A supply center for solvents.

The self-service shops are run by Thermo Fisher Scientific, Nordic Biolabs and Qiagen. The supply centers are set up as online shopping systems; after registering a purchase from the supply center, the customer picks up the products directly from the shelves. Refills for the purchase will be sent automatically. An introduction to the purchase systems is required, and a customer contract needs to be signed and handed over to Site Support in order to get access. Please note that access to the centers is personal, and so sharing accounts is not allowed.

Please note that all universities have different framework agreements. Please make sure you follow the framework agreement of your home university when purchasing products from the supply centers.

More information on the supply centers and on how to become a user here: intranet.scilifelab.se/lab-orders/supplycenter

3.9 Personal Protective Equipment (PPE)

The reference for this procedure is AFS 2023:11, 15 chap. on the choose and use of personal protective equipment.

Personal protective equipment (PPE) is to be used at all times in the laboratory in order to protect against

- Chemical spills on hands and clothes.
- Eye contact with hazardous or corrosive liquids and solids.
- Inhalation of toxic or irritating aerosols.
- Radiation (ionizing or UV).

PPE is also used to protect your experiments from contamination or other interference.

Lab coats should be worn at all times in the laboratory and should be removed when leaving the lab and entering common office areas or corridors. Dirty coats should be dropped in the laundry basket in environmental rooms. Heavily soiled or contaminated lab coats should be disposed of in the yellow waste bins for laboratory waste.

Appropriate protective gloves should always be used when working with

- Chemicals. Regardless of concentration, chemicals should always be considered toxic.
- UV light or hot fumes (steam/ånga).
- Sharps or cutting objects.
- Bacteria, cell cultures, viruses, GMM, blood/tissues.
- Cold substances at low enough temperatures to cause damage, such as liquid nitrogen and dry ice. Extra padded gloves should be used.

Gloves should be *removed* when

- Leaving the lab and before touching any door handles!
- Using computers or other equipment such as keyboards or dials.
- Working with an open flame.

Protective goggles/glasses should always be worn when working with the following materials outside of a shielded environment (i.e. fume hoods or safety cabinets)

- Chemicals. Regardless of concentration, chemicals should always be considered toxic.
- UV light or hot fumes (steam/ånga).
- Sharps or cutting objects.
- Bacteria, cell cultures, viruses, GMM, blood/tissues.
- Working with cold substances at low enough temperatures to cause damage, such as liquid nitrogen and dry ice.

Please note

- The use of contact lenses will *increase* the risk of eye injury if exposure occurs.
- Use safety glasses with *full cover* when working with toxic/corrosive chemicals, standard laboratory glasses cannot provide full protection.

Respiratory protection masks should always be worn when working with chemicals that can have an impact on your respiratory tract, and/or that create a lot of aerosols. Respiratory masks should also be used when handling virus, blood or human tissue samples where there is risk of aerosols or splashes, and always when handled outside of a shielded environment.

3.10 Risk Assessments

Before initiating any activities involving hazardous materials or risky operations, a comprehensive risk assessment must be conducted. This is crucial for ensuring safety (for yourself, others, and the environment) and compliance with legal requirements.

A standard risk assessment relies on information from the Material Safety Data Sheet (MSDS) of the product and instruction manuals, covering the following areas:

- Experimental Design and Procedures for identifying associated risks including when they happen, who is affected and how.
- Preventive and Protective Measures
- Accident Response Procedures
- Waste Disposal Plan
- Final Risk Evaluation

Templates are readily available to facilitate risk evaluations for common laboratory hazards: chemicals (KLARA), human material (HUMRA) and biological agents (BARA).

All risk assessments must be overseen by the individual responsible for the work environment (Principal Investigator, group leader, etc.). Additionally, a copy of each risk assessment should be easily accessible within the laboratory. A copy of HUMRA and BARA must also be uploaded to a cloud service which is accessible to SciLifeLab Campus Solna administration. Contact your Safety representative for more information.

A **HUMRA** (Human Sample Material Risk Assessment) or similar form should be filled out together with a supervisor prior to working with *human sample material and blood (including primary cell cultures/tissues).*

A **BARA** (Biological Agents Risk Assessment) or similar form should be filled out in order to assess the risk of handling *microorganisms* (including virus and cell cultures). The BARA is also used for ANIMAL risk assessments.

Find HUMRA and BARA risk assessment templates on the intranet: *Working Environment and Lab Safety>Biosafety at Campus Solna.*

4. Common SciLifeLab Licenses and Permits

Site Support administers the following laboratory licenses and permits for all groups located at SciLifeLab Campus Solna;

- Permit for Flammable Products (Tillstånd Brandfarlig Vara)

  The permit for keeping and handling flammable liquids and/or gases at SciLifeLab Campus Solna is administered by the Site Support. Handling of flammables at SciLifeLab

Campus Solna is therefore strictly reliant on the permit. All purchases of flammable gas (>0L) and all larger volumes/storages of flammable liquids (>50L) must be reported to Site Support and the Lab Safety Coordinator for permit purposes. Please also inform the Site Support prior to purchasing larger volumes of gas under pressure. *For more information, see section 6. Gas and Flammables*.

- Genetically Modified Microorganisms (GMM) - Level F

  Site Support registers all activities at SciLifeLab Campus Solna related to genetic modified microorganisms at biosafety level 1 (GMM-F, negligible risk), with the Swedish Work Environment Authority (Arbetsmiljöverket). For license- and safety reasons all new GMM-F activities must be reported to the SciLifeLab Lab Safety Coordinator indicating the following:

  - GMM-F's species
  - Waste treatment
  - Room where you intend to work with GMM-F

  For more information on licenses for GMM and biosafety see section 7. Biosafety.

Additional permits or licenses are required when handling for example: A/B class chemicals, GMM and other microorganisms at level L (such as virus), GMO, radioactive chemicals, and drugs/narcotics. These are *not* registered by Site Support. However, SciLifeLab Lab Safety Coordinator must be informed prior to starting work that requires additional permits or licenses. **For license- and safety reasons all new permits or licenses (regardless of safety level) have to be reported to the SciLifeLab Lab Safety Coordinator.** *For more information on permits, visit your home university staff pages or contact SciLifeLab Lab Safety Coordinator.*

5. Chemical Safety

Staff at SciLifeLab must adhere to both their home university's chemical handling regulations and the common rules established at SciLifeLab.

The following are common tasks related to chemical handling delegated to group leaders at SciLifeLab CS:

1. Ensure all necessary permissions for chemical handling are in place.
2. Develop accurate risk assessments for handling any chemical that could pose a risk.
3. Ensure that individuals handling chemicals have been properly introduced to the operations and possess sufficient knowledge for safe handling.
4. Ensure appropriate equipment is available for handling the chemical. This includes safety equipment, material for transport and spill cleanup.
5. Ensure proper storage of the chemical and its registration in the KLARA database.

5.1. KLARA Chemical Management System

According to Swedish work environment laws, including AFS 2023:10, 7-10 chap. on chemical risk sources, all organizations and companies must maintain a registry of hazardous chemicals. KI, KTH and SU have all implemented the KLARA chemical management system for documenting all hazardous and non-hazardous laboratory chemicals (excluding medicines and drugs). Research groups and platforms at SciLifeLab are required to register their chemicals at their respective universities.

Functions of KLARA

1. **Chemical Search:** Enables users to search for chemicals to check stock levels and storage locations.
2. **Safety Information:** Provides access to safety data sheets and other safety information for chemical products.
3. **Risk Assessment:** Allows users to conduct risk assessments for chemicals.
4. **List and Report Creation:** Facilitates the creation of lists and reports, such as those for high-risk chemicals or flammables.
5. **Inventory Management:** Enables the maintenance of a chemical register and performs inventories (restricted to inventory takers).

Access to KLARA

All personnel handling chemicals should have access to KLARA. KLARA accounts are personal and non-transferable. Users access their accounts through their university's KLARA platform, utilizing their university ID for authentication. Subsequently, they should request a KLARA administrator to link their account to their respective group, providing the group name and specifying the type of access required (reading and risk assessments/inventory taker). You can find information about your current KLARA administrator and access the KLARA platform through the intranet.

Users at SciLifeLab who require access to KLARA at another university can do so through their home university's KLARA platform. They simply need to log in with their university credentials and then follow the link to the specific KLARA they wish to visit.

Annual Chemical Inventory

At SciLifeLab, each research group or platform that owns chemicals appoints a KLARA representative or inventory taker. The primary responsibility of the KLARA representative is to conduct the yearly inventory, ensuring the accuracy and currency of the chemical registers.

Barcode Module

The barcode module streamlines inventory processes, facilitating the addition of newly purchased chemicals and removal of finished chemicals throughout the year. Regular updates simplify chemical location and help prevent unnecessary purchases.

KLARA barcode equipment (scanner, label printer, and laptop) is available for borrowing from Site Support. Users can book equipment via the instrument booking system and collect it from the reception.

5.2. Safety Data Sheets (SDS)

By law, all suppliers of hazardous chemical products must provide purchasers with a Safety Data Sheet (SDS), which contains crucial information about the chemical. The SDS information must be up-to-date and according to the UE regulation. The SDS is structured into 16 mandatory sections, covering topics such as hazard identification, first-aid measures, appropriated personal protective equipment (PPE), waste handling, and exposure controls.

SDS are accessible through the KLARA chemical database and online resources. It is essential for all employees to be proficient in locating and utilizing SDS to ensure the safe planning and execution of experiments.

5.3. Chemical risk assessment (KLARA)

Ensure you understand the risks associated with any chemical before handling it. Always conduct a risk assessment before beginning your experiments. Risk assessments within KLARA assist chemical users in meeting the requirements outlined by the Swedish Work Environment Authority in their provision AFS 2023:10, 7-10 chap. on chemical risk sources. A copy of the assessment is securely stored within the system. Use Safety Data Sheets for guidance and safety precautions.

To create a new assessment in KLARA, log in to KLARA and navigate to the "Chemicals" section, then select "Risk Assessments". Click on "New Assessment" located in the top left frame. Once initiated, you will encounter various fields that need to be completed. Each field includes helpful text explanations that can be accessed by clicking on the blue “i” icon. After filling out all the necessary information, click “save” and change the status to "finished". Notify your supervisor so they can review and approve it.

5.4. Chemical Handling

As mentioned earlier, a risk assessment must always precede any chemical work. Below are some general safety procedures for working with chemicals:

- Minimize the volume/amount of chemical you are using or transporting.
- All solutions intended for storage or long-term use must be labelled at least with content, relevant CLP hazard pictograms and text with information thereon when the product is carcinogenic, allergenic, damaging to heritable genetic material or toxic to reproduction.

- Use a fume hood when handling chemicals that could pose a risk of harmful exposure. Work with the sash as far down as possible using forced airflow. Be sure to remove chemicals and solutions after the work is finished. The fume hoods should be clean and tidy; they are not storage locations.
- Chemicals should be properly sealed, labeled, and returned to the appropriate storage area after use.
- Follow SciLifeLab regulations for appropriate waste disposal (*See section 10. Hazardous waste)*.

Internal transport;

- Do not transport full bottles of chemicals around the lab without proper protection. Use an appropriate container, box or rack for safe transport.
- When transporting dangerous goods within the building, an extended risk assessment should be carried out, taking the following into account:
    - Avoid passing through areas with a high flow of people or where a lot of people are staying
    - Minimize the amount of chemical products being transported.
    - Adapt the route and time when transporting large quantities.
    - Personal protective equipment must not be worn in public areas. This should not be needed as transport packaging must be clean from contamination on the outside.
    - The material must be packed so that leakage / spillage is prevented even if the packaging is dropped.
- The transport of dangerous goods (all chemicals with a UN number) should be done in the goods elevator. Never transport dangerous goods in the personnel elevator. When utilizing the goods elevator for chemical transport, ensure that the SciLifeLab Hazard Sign is displayed.
- An elevator transport key can be used to operate the elevator remotely, allowing it to travel between floors without personnel inside. This feature also prevents the elevator from stopping at any intermediate floors. Consider using the transport key when transporting large quantities of chemicals, waste, pressurized bottles, or if your risk assessment indicates that it is necessary. During liquid nitrogen transportation via elevator, no personnel are permitted to accompany the container.

Storage of chemicals;

- Store chemicals according to their inherent properties, as described by the SDS.
- All cabinets, cupboards, freezers, fridges in which chemicals are stored must be labelled with the relevant hazard pictograms.
- Chemical products shall not be stored so that there is a risk that they accidentally end up in the sewage (e. g. in a fume hood without embankment, on shelves above a sink)
- Different types of hazardous chemicals should not be mixed. Keep separate and ventilated storages for acids, bases, toxics and flammables.
- CMR substances and poisons should be kept separately and not be easily available for all.
- Corrosive chemicals (acids and bases) should be stored

- Oxidizing chemicals should not be stored with oxidizable substance.
- Peroxide-forming chemicals should be stored in a dark and cool place.
- For storage conditions for flammables gases and liquids, please see section 6.3.

5.5. Chemicals subject to authorization

Certain chemicals require authorization from a Swedish Authority or are subject to additional regulations. It is the responsibility of the group leader to ensure all necessary permissions are in place before placing an order. Within KLARA, you can access reference chemical lists that will assist you in identifying chemicals with specific requirements.

The following types of chemicals are prohibited or require special permission:

1. Group A substances, use prohibited (Swedish work Environment Authority)
2. Group B substances (Swedish work Environment Authority)
3. Goods dangerous to health (Public Health Agency of Sweden)
4. Mercury and Mercury compounds (Swedish Chemicals Agency)
5. Narcotic precursors, Medical products Agency category 1 (Swedish Medical products Agency)

Questions about permits can be forwarded to SciLifeLab Safety Coordinator or to the person responsible at your home university. Please note that work with these substances must be reported to SciLifeLab administration.

5.6. Chemicals subject to special regulations and handling

Apart from chemicals that require a permit, there are various types of chemicals that have specific requirements. For instance, certain educational qualifications, risk assessments, or medical examinations may be necessary before beginning work with a particular chemical.

You can use the Chemical database in KLARA to find out chemicals subject to specific regulations as well as links to these regulations. To access the database, log in to KLARA and navigate to the Chemical Database. You can search chemicals by name or CAS number. Examples of types of chemicals subject to specific regulation within KLARA include:

- Sensitising Substances
- CMR substances
- Diisocyanates
- Chemicals that may affect pregnant and breastfeeding workers.
- Chemicals for which medical examination is required or must be offered.
- Perchloric acid
- Potential peroxide forming chemicals.

Additionally, you can search among your registered chemicals by using the “Search Product” function. Simply select the desired chemical list from the “Only Include Products Listed As” menu to refine your search.

5.7. CMR Chemicals

CMRs (carcinogenic, mutagenic, or toxic to reproduction substances) are highly toxic chemicals that can cause severe health effects upon exposure. CMR classified products are labelled with the hazard pictogram for “Health Hazard” and with the following hazard statements:

- H350: May cause cancer.
- H340: May cause genetic defects.
- H360: May damage fertility or the unborn child

Handling of CMR chemicals is subject to special regulations:

A CMR chemical must be substituted with a less harmful alternative when feasible. Therefore, before working with or ordering CMR chemicals, a CMR investigation to explore substitution possibilities must always be conducted. KLARA provides a tool that facilitates this investigation and documentation as part of your chemical risk assessment. Handling a CMR product is allowed only when such documented investigation demonstrates that switching to a less hazardous alternative is not technically feasible (AFS 2023:10, 8 chap. §14).

Before handling CMR chemicals always conduct a thorough risk assessment. The assessment should clearly identify where the substance may occur and what measures should be taken to ensure that only necessary personnel are present, the safety measures necessary to ensure minimal exposure, when and which personal protective equipment must be used, and how and when equipment, processes or ventilation monitoring should be conducted to early detect deviations that could pose increased risk (AFS 2023:10, 8 chap. §15).

CMR exposure register:

If there is a risk that people have been exposed to a CMR substance due to e.g. accident, spillage, ventilation interruption, non-compliance with routines/instructions, failure to assess risk, the employer must establish a register of the persons who have been exposed. This register must be kept at your university for at least 40 years, except for products labelled with the hazard statement H360, which must be kept for at least 5 years (AFS 2011:19, §41). The aim of the register is to facilitate investigations of work-related diseases.

5.8. Allergenic chemical products

The handling of certain allergenic substances is subject to specific regulations as outlined in AFS 2023:10, 8 chap. These substances include:

- chemical products labelled with H317 (may cause allergic skin reaction) or H334 (may cause asthma symptoms or breathing difficulties).
- chemical products which contain ethyl- or methyl-2-cyanoacrylate.
- work that entails thermal degradation of materials that release iso-cyanates or processes that release formaldehyde.

To determine if this regulation applies to your chemical, you can use the Chemical Database in KLARA. In such cases, specific training or medical check-ups may be mandatory. Failure to comply could result in penalty fees.

5.9. Dry Ice and Liquid Nitrogen

When working with liquid nitrogen/dry ice there is a risk for frostbite, suffocation or explosion. Work involving liquid nitrogen and dry ice may only be carried out by those who have adequate knowledge of the chemical and the potential risks that handling and use entail, together with how these risks can be avoided.

It is the duty of the group leader to ensure the following:

- A thorough risk assessment has been conducted before beginning work with liquid nitrogen/dry ice.
- Written handling and safety instructions are established and accessible.
- Adequate knowledge of the associated risks and safe handling practices is understood by all affected personnel.

General safety measures:

- Always wear the appropriate PPE, including a face visor/protective goggles, full-cover shoes, padded gloves, and a lab coat.
- Utilize dry ice and liquid nitrogen only in well-ventilated areas.
- Never work alone with liquid nitrogen.
- Use designated liquid nitrogen containers for transportation and storage. For dry ice, it is recommended to use a styrofoam box or cooler bag. Never use "coffee thermoses" or similar containers due to the risk of explosion.
- Never transport yourself or others with liquid nitrogen in elevators, as even a minor spill can lead to dangerously low oxygen levels in enclosed spaces.

In Case of Spill:

- In the event of a small spill, evacuate the area until the liquid nitrogen has completely evaporated. For larger spills or instances of poor ventilation, seal off the affected area. Consider implementing measures to enhance ventilation if necessary. If there's a risk of compromised oxygen levels on your floor, activate the floor evacuation alarm and promptly contact Site Support or chemical experts for assistance. (Refer to section 9.1 Emergency Contacts in Case of Fire, Evacuation, and Accidents).

- For dry ice spills, use a brush and dustpan to sweep up the pellets. Ensure proper ventilation in smaller spaces. If uncomfortable handling the spill, contact the nearest Safety representative or Site Support for assistance.

Please, refer to the document "**Basic of liquid nitrogen handling at SciLifeLab"** for further information (*Intranet> Work environment and lab safety> Chemical handling*)

Services at SciLifeLab Campus Solna

Dry ice can be picked up from the dry ice station in Delta floor 1 by corridor to the gamma building.

Refill and transport of **liquid nitrogen** in approved containers to your labs is handled as a central service. Please plan your lab work to take this routine into account.

- You place your empty container outside the goods/ loading dock in Delta 1 (inside the marked area) before 10.00 to receive the delivery before 14.00 the same day.

- Label the canister with the following information:

  * Name

  * Phone number

  * How much liquid nitrogen you need

  * Room number / location for delivery

6. Gas and Flammable Products

6.1 Gas

Large, pressurized gas tubes must be stored standing up and always be anchored to a wall. Transportation of gas tubes requires using a special gas-trolley where the gas tube is anchored to the trolley.

All new purchases of flammable gas must be reported to the SciLifeLab Lab Safety Coordinator prior to the purchase due to license and safety reasons (this is not required for refills).

6.2 Central Gas Storage Room A1811

The gas storage room (A1811) is the SciLifeLab central storage for compressed inert gases e.g. Nitrogen, Helium, Argon, Carbon Dioxide and Oxygen.

Special gases with flammable properties (Propane/butane, methane, ethane) are stored in the separate EI60 class room, A1830. This room is fireproof and self-ventilated.

Leakage of gases other than oxygen can lead to oxygen deficiency by displacing the oxygen in the room. The gas storage room is therefore equipped with an oxygen level alarm. The oxygen level alarm is local and is not connected to the emergency services. Please consult Section 8.8 regarding the low oxygen alarm for a detailed description and instructions on how to respond in the event of an alarm.

Please be advised that access to the gas storage room is restricted to authorized personnel only. Individuals requiring entry must be introduced to safety protocols, equipment operation, potential hazards, and familiarization with the oxygen alarm system before being granted access. Students and new employees are strictly prohibited from entering the gas storage room without supervision. For further inquiries, please contact Site Support.

6.3 General Handling and Storage of Flammable Products

- Handled volumes of flammable gas and liquid must always be minimized.
- Flammable products must be stored in fireproof ventilated cabinets (minimum EI60 class). Smaller amounts (< 1 L) of flammable liquids such as Ethanol, Propanol or Methanol can be stored in regular ventilated cabinets in the lab until a maximum of 50 L in total per floor.
- Only quantities necessary for immediate use should be removed from storage during working hours and returned to the storage by the end of the day.
- Flammable substances should be stored separately and must not be stored with toxic, corrosive or oxidizing chemicals.
- Halogenated substances such as dichloromethane and chloroform cannot be stored with flammable substances.
- Different types of flammable substances (i.e., gas, liquids, reactive) should be stored separately.
- Combustible materials are strictly prohibited in storage locations for flammables.
- Flammable substances must not be stored in standard refrigerators or freezers. Instead, ensure the use of spark-proof and EX-class refrigerators and freezers for safe storage. Please note that storing hydrogen peroxide in the same refrigerator is not allowed. Furthermore, it is not permitted to place any refrigerator or freezer inside an EX-classed area.
- The application of flammable surface disinfectants such as 70% ethanol with a spray bottle is not permitted. Instead, squeeze wash bottles are recommended. In exceptional cases and following a risk assessment, a spray bottle may be approved for specific tasks inside cell labs or BSL2 labs where there is a high risk of material contamination.

7. Biosafety

Laboratory biosafety embraces facilities, equipment, practices and procedures deemed to reduce or prevent the risk of exposure of people and environment to contagious biological material generated in the laboratory.

Biological risks or biorisks are risks associated with biological material or infectious agents. Examples include toxins, prions, viruses, bacteria, fungi, parasites, cell cultures, blood and tissues, research animals, plants and contaminated laboratory waste.

7.1. Animal Models

There are no animal facilities at SciLifeLab. Work with live animals is therefore not permitted. Please note that animal experiments require ethical permission and are subject to additional laws and regulations. *For more information on animal models, contact the animal facility at your home university.*

7.2. Microorganisms

References for the following biosafety routines are the Work Environment Authority's provisions AFS 2023:10, 11 chap. (*Smittrisker* in Swedish) and the AFS 2023:13, 9 chap. *Contained Use of Genetically modified Microorganisms*.

- **Microorganism:** A microorganism is defined as "every microbiological unit, cellular or non-cellular, which is able to reproduce, or transfer genetic material". They include, but are not limited to, bacteria, viruses, yeast, protozoa and cell cultures of higher organisms.

- **GMM (Genetically modified microorganism):** GMM is defined as a microorganism, in which the genetic material has been altered in a manner that would not occur naturally through mating or natural recombination.

Microorganisms are classified into risk groups 1-4. Microorganisms that cause infectious diseases in humans are classified, in accordance with increasing hazardousness, into risk groups 2-4. The ones that do not cause infectious diseases are classified as risk group 1. They may still cause ill-health due to hypersensitivity or toxigenic effects. Examples of microorganisms that do not normally cause infectious diseases are cell cultures, as well as *E. coli* which has been adapted to the laboratory (i.e., not pathogenic).

You can find a list of microorganisms with their risk group in the Swedish Work Environment Authority's provision AFS 2023:10, appendix 7.

Biosafety or containment level 1-4 often corresponds to the risk group of the microorganism but takes into consideration specific procedures used that may alter exposure level and required control measures. Therefore, **the risk assessment regulates which biosafety level (BSL) must be applied**. For example, an organism that is classified into risk group 2, may be handled in a BSL 3 lab if high concentrations of aerosols are generated in procedures used. In the case of GMM, apart from microorganism risk group and procedures, type of vector and insert need to be taken into consideration.

7.2.1. Permits and notifications

Group leaders are responsible for checking that required notifications are in place for their work.

Notifications and/or permit from the Swedish Work Environment Authority are required before beginning work with wildtype microorganisms belonging to risk group 2 or higher and GMM of all risk classes. ***Biosafety level 3 and level 4 work is NOT allowed at SciLifeLab Campus Solna***.

GMM-F notification: Site Support administers the general GMM level 1 (GMM-F) notification at SciLifeLab Campus Solna. If you are going to work with GMM belonging to risk group 1, you need to make sure that the rooms you work in are part of the notification. A list of rooms included in the SciLifeLab Campus Solna GMM-F notification is available on the intranet. All GMM-F activities at new locations at SciLifeLab must be reported to the SciLifeLab Lab Safety Coordinator, who will update the GMM-F notification. Please, indicate the following information:

- Risk group 1 GMM species
- Waste treatment
- Room where you intend to work with GMM

Notification of work with microorganisms in risk group 2 and GMM-L activities (GMM level 2): Before starting the work, the SciLifeLab Lab Safety Coordinator and/or a home university biosafety coordinator/delegate must be contacted and the work approved. Biosafety delegates may provide comments and submit the report to the Swedish Work Environment Authority. A copy of the notification must be sent to SciLifeLab Lab Safety Coordinator.

7.2.2. Risk assessments

A risk assessment shall always precede all experiments involving biorisks.

BARA (Biological Agents Risk Assessment) form can be used to assess the risk of handling microorganisms (including cell cultures) that are not genetically modified. The form is available on the intranet.

Work with GMM on safety level 2 requires a risk assessment made on the Swedish Work Environment Authority's form for notification of GMM-L activities.

Work with GMM on safety level 1 requires a risk assessment made on the Swedish Work Environment Authority's form for notification of GMM-F activities. The risk assessments do not have to be reported to the Swedish Work Environment Authority.

The risk assessments shall be made available in connection to the premises where the experiments are carried out, e.g., in a laboratory binder or similar. This shall be updated in the event of any changes, and be available to display at any safety rounds, and inspections by the Swedish Work Environment Authority.

7.3. Human blood, tissues and cell cultures

Human blood includes blood products and material contaminated with blood such as tissues, blood plasma, spinal fluid, sputum, tumors, and primary cell cultures. It also includes contaminated laboratory material such as pipette tips, gloves, cannulas, and other consumables.

The handling of human blood is regulated in the Swedish Work Environment Authority's provision AFS 2023:10, 11 chap. *(Smittrisker* in Swedish). It does not require a specific permit provided that no microorganisms are propagated in the sample.

The main rule for blood handling is to **handle all blood and blood contaminated material as if it was infectious.** It is primarily Hepatitis B and C as well as HIV that can be transmitted through human blood handling.

The laboratory must be at least biosafety level 2 (**BSL2 lab**), must have access to skin disinfectants and there must be protection against contamination from splashes. There must be written instructions on how to handle blood, which shall be based on a risk assessment.

Blood handling may be carried out only by those who have received **training**. Personnel are offered hepatitis B **vaccination** and /or test to check for protection.

Please, see the following documents on the intranet:

- Guidelines for working with human blood, tissues and cell cultures
- Risk assessment form for blood and other human sample materials – HUMRA.

7.4. Animal Tissues

Please note that handling tissues from animals might be subjected to the same safety regulations as human blood and tissue. Sometimes animals and/or animal tissues can carry infections you are not aware of, an example being the Herpes B virus in apes. Infection with Herpes B virus in humans is very rare but can lead to serious illness. The infectious diseases that can be transmitted from animals to humans are called zoonoses. You can find more information on this on the work environment authority webpages. All animal tissue samples, regardless if they originate from infected animals or healthy, should be considered infectious.

Always risk assess experiments with animal tissues prior to the handling. Technical equipment might be required such as splash protection or safety cabinets depending on the assessed risk.

7.5. Other Biorisks

7.5.1. Genetic Modified Organisms (GMO)

This section refers to organisms, non-microorganisms, in which the genetic material (DNA) has been altered in a way that does not occur naturally by mating or natural recombination. GMOs include, for example, plants, seeds or animals whose genetic material has been altered using genetic engineering techniques.

Permission from specific Swedish Authorities is required for work with GMO. Please, contact SciLifeLab Lab Safety Coordinator before introducing a GMO to SciLifeLab Campus Solna.

7.5.2. Animal Bi-products

Permission from the Swedish Agricultural Agency (Jordbruksverket) is required for work with animal bi-products. Contact SciLifeLab Campus Solna Lab safety coordinator for more information.

7.5.3. Invasive species

Permissions may also be required for work with invasive species or the importation of organisms. Invasive species have the potential to cause great damage to the environment. Before importing any organisms, check if special permission or containment is required. Contact SciLifeLab Campus Solna Lab safety coordinator for more information.

7.6. Safety Routines

7.6.1. Biosafety Level 1

Biosafety level 1, the lowest level, applies to work with biological agents that usually pose a minimal potential threat to laboratory workers and the environment and do not at all or not consistently cause disease in healthy adults. Research with these agents is generally performed on standard open laboratory benches without the use of special containment equipment. BSL 1 labs do not have to be isolated from the general building but areas where GMMs are handled should be clearly marked if in an open lab space shared with external groups. Training in the specific procedures should be given to the lab personnel.

Good Laboratory Practices should always be followed, splashes and aerosols should be avoided, sharps should be handled carefully, and all work surfaces should be carefully decontaminated when the work is finished. Hands should be thoroughly washed and disinfected after handling biorisks. Proper PPE should be used, and waste should be handled according to SciLifeLab waste handling routines (*See section 10. Hazardous Waste*). In the case of GMM-F, although they do not involve risks to human health or the environment, they must be kept indelible. This means that they cannot be taken out of the facility via sewage or regular waste.

7.6.2. Biosafety Level 2

Biosafety level 2 (BSL2) covers work with agents associated with human disease, i.e. pathogenic or infectious organisms posing a moderate hazard. Infection can lead to diseases of different gravity, but the infections can normally be cured, be prevented or self-heal. Examples of safety class 2 biological agents are *Listeria ivanovii, Salmonella typhimurium*, *Streptococcus spp*. *Adenoviridae*, Human B-lymphotropic virus (HBLV-HHV6), Human papillomavirus, Human parvovirus (B 19). BSL2 is also required for work with human blood (see section 8.2)

The laboratory should fulfil all the **requirements** for a BSL 2 lab. A complete list of requirements for BSL2 labs at SciLifeLab can be found on the intranet.

There must be **written instructions** on how to handle the biorisks specific for each lab space. The written instructions should be based on a **risk assessment** of what biorisks are present within that work space.

Work may be carried out only by those who have received **training**. The training must include information on common routes of exposure, protective measures, accident response, as well as good microbiological practices and workplace hygiene.

Please, see the following documents on the intranet:

- Guidelines for working with human blood, tissues and cell cultures
- Guidelines for working with Biological Safety Level 2 (BSL2) including GMM-L Biorisks.
- BARA and HUMRA

7.6.3. Biosafety Level 3 and 4

Note: Biosafety level 3 and level 4 work is NOT allowed at SciLifeLab Campus Solna.

Level 3: Yellow fever, St. Louis encephalitis, HIV, and West Nile virus are examples of agents requiring biosafety level 3 practices and containment. Work with these agents is strictly controlled and requires license. These are indigenous or exotic agents that may cause serious or lethal disease via aerosol transmission, i.e., simple inhalation of particles or droplets.

Level 4: Agents requiring BSL 4 facilities and practices are extremely dangerous and pose a high risk of life-threatening disease. Examples are the Ebola virus, the Lassa virus, and any agent with unknown risks of pathogenicity and transmission. These facilities have to provide the maximum protection and containment.

7.6.4. Protective Measures

The protective measures that should be chosen for the handling of biorisks depend on the way the material transmits infection. Take into consideration that chemicals and biological material are often handled simultaneously. The choice of protective equipment should be defined during the risk assessment.

Gloves

The risk assessment defines whether gloves need to be used, and if so, which type of gloves. When choosing gloves for microbiological work, both penetration ability and resistance to cut damage must be considered. Your hands shall always be washed with water and soap and disinfected upon taking off the gloves after microbiological work. Remember that the disinfectant destroys the protective properties of the glove, change them regularly.

Protective clothes

A lab coat should always be worn when carrying out microbiological work. The clothing shall be removed when leaving the work area.

Internal transport

When transporting microorganisms between floors at SciLifeLab Campus Solna, samples should be properly packed in a clean box or similar and protective clothing removed. Personal protective equipment should not be needed as transport packaging must be clean from contamination on the outside. The material must be packed so that leakage / spillage is prevented even if the packaging is dropped.

Ventilated workstations and splash protection

The risk assessment for the experiment shall state whether ventilated work areas need to be used, and which type. It is important to differentiate between microbiological safety cabinets (also known as biological safety cabinets, BSC) and ventilated workstations adapted to handling chemicals.

A **BSC** is adapted for work with microorganisms. Among other things, it has an HEPA filter, in order to purify the outgoing air. The majority of the air circulates, most often but not always, inside the BSC, and HEPA filtered air blows from the inner ceiling of the BSC (this is called a class II BSC). This is important to not use a ventilated workstation where the air flow is directed from the material to the worker. This is normally called laminar flow cabinet and only protects the sample from contamination, but not the worker.

A **ventilated workstation adapted to handling chemicals** (e.g., down flow bench, local extraction system and fume hoods) has no HEPA filter, but may have, e.g., a coal filter, which does not protect against microorganisms in the air. Ventilated workstations adapted to handling chemicals do not, generally, protect the environment against microorganisms.

It is also important to know if the ventilated workstation is connected to the building ventilation system or not. A BSC is normally not connected to the ventilation and therefore it is only suitable for microbiological work, not for chemicals. A BSC connected to the building ventilation can be used for combined work with biorisks and small amounts of chemicals.

When only **splash protection** is needed, as is often the case with blood handling, a Plexiglas sheet, visor or safety goggles and facemask can be used. In these cases, a fume hood or BSC also works well for splash protection.

Vaccination

When working with specific microorganisms and when doing work that entails a risk of blood infection, for which there is a vaccine, personnel shall be offered vaccination. Vaccinations can be provided by your home university health care. *For more information, see section 2.3 Occupational Health Care.*

Note that while risks associated with exposure to blood and tissues contaminated by Hepatitis B can be mitigated by vaccination, the potential risk of infection by other agents such as Hepatitis C and HIV and others, can only be reduced by following good laboratory practices when handling specimens. Always handle specimens in appropriate laboratory facilities and according to the lab specific handling instructions/risk assessments, and follow all safety precautions as outlined in this document.

7.6.5. In case of accident

A spill plan and spill kit should be available in all labs where biorisk are present. If a spill is too large to manage, close off the space and seek help. Contact your closest Safety representative and report the incident.

Please, see the following document on the intranet:

- Accident Response Plan for Laboratories Working with Biorisks

8 Fire safety and Emergency Procedures

8.1 Emergency Contacts in Case of Fire, Evacuation and Accidents

SOS EMERGENCY AMBULANCE, FIRE BRIGADE, POLICE	**112**
ALARM Report special incidents such as serious accidents, threats or fire (after you have called 112). “KTH emergency number” but works for all universities. State “SciLifeLab” in the errand.	08-790 7700

SECURITY SUPPORT 24 h number for non-urgent threats, sabotage, burglary. “KTH security support” but works for all universities. State “SciLifeLab” in the errand.	08-790 9900
SWEDISH POISON INFORMATION CENTRE Non-emergency	08-33 12 31
GENERAL HEALTH ADVICE *Vårdguiden* (on-call health care advice)	www.1177.se or 1177

HOUSE FAILURES Failures regarding electricity, heat, ventilation, water leakage. AKADEMISKA HUS, JOUR (15.30-7.00)	010-557 24 00

CHEMICAL SPILL (Advice, help with cleaning-up)

SEKA – Office hours (8.00-16.00)	08- 23 53 00

GENERAL SCILIFELAB LAB SAFETY CONTACT	
Lab Safety Coordinator (Office hours 9.00-16.00)	08- 790 64 57
Site Support	support@scilifelab.se

8.2 General fire safety

Lithium batteries

The use of e-bikes and e-scooters equipped with lithium batteries is an environmentally friendly and easy way of transportation and is becoming more and more popular.

Within SciLifeLab Campus Solna premises, it is not allowed to charge lithium batteries due to fire hazards, nor to store your bike/scooter indoors.

Bike parking facilities are available outdoors, in front of the main entrance, and indoors, in a locked bike room within the garage Skruven (Tomtebodavägen 5B).

Electrical household appliances

Private electrical household appliances must not be used within SciLifeLab Campus Solna premises. Necessary electrical household equipment such as kettles must be CE marked and purchased through your university framework agreement.

Electrical household appliances may not be used in offices or laboratories, but only in rooms intended for dining and rest- areas such as lunch and pantry rooms.

Coffee machines/kettles in kitchens may only be connected to electrical outlets fitted with an electronic timer built into the electrical outlet. The exception to the requirement for a timer is the coffee machines supplied by SciLifeLab CS Site Support.

Heating equipment

Temporary heating equipment (e.g. portable electric radiators) should generally not be used, since they can disrupt the building’s ventilation system and hurt energy consumption. Use should be preceded by dialogue with SciLifeLab CS Site Support.

When portable radiators must be used, CE-marked oil-filled portable radiators must primarily be purchased through your university framework agreement. Great care must be taken so that the radiator does not risk being covered in any way, and that there is sufficient free space around the radiator for air to circulate freely. Combustible material must not risk coming in contact with the radiator.

Portable radiators must be switched off when the working day is over.

Storage on top of fridges and freezers

Storing combustible materials, such as cardboard boxes, on top of fridges and freezers is strictly prohibited, even if the appliances are spark-free.

8.3 Fire safety organisation

The SciLifeLab Campus Solna fire safety organization consists of one fire safety supervisor (Sw: *brandskyddsansvarig*), one manager of flammable goods (Sw: *föreståndare av brandfarlig vara*), and appointed evacuation leaders, inspectors SBA and inspectors of flammable substances. An updated list of responsible people can be found on the intranet: https://intranet.scilifelab.se/working-at-scilifelab/for-your-work/responsibility-areas/

- Inspector SBA

The SBA inspectors (Inspektör Systematiskt Brandskydds Arbete) at SciLifeLab inspect the fire safety equipment and the general fire safety level on the floor at regular intervals and reports any faults to the SciLifeLab Fire Safety Supervisor. There is at least one SBA inspector on each floor.

- Inspector Flammable Substances

The inspectors of flammable substances ensure that flammable substances are stored according to the SciLifeLab regulations and forward any questions or issues to Site Support.

- Evacuation stations and leaders

Each floor has two or more evacuation stations, where the orange evacuation leader's vest and instruction card are stored. Additionally, each floor has at least two appointed evacuation leaders. However, they may not always be on site when the fire/evacuation alarm is activated. If the designated leaders are absent, the first person to arrive at an evacuation station is expected to take on the role of evacuation leader. Put on the vest and follow the instructions on the card.

The evacuation leaders' primary task is to order an immediate evacuation and guide people to an appropriate escape route. **Their first priority is always their own safety.** Once it is safe to do so, the evacuation leaders check the floor, including toilets and changing rooms, before leaving. They then report to the SciLifeLab fire safety supervisor (yellow vest) or to the leader of the rescue team, providing relevant information to the best of their knowledge, for example, whether anyone remains in the building and the reason for the evacuation.

It is everyone's responsibility to know the locations of the evacuation stations and to familiarize themselves with the information on the instructions on the card.

- Fire Safety supervisor

The Fire safety supervisor shall organize, document, inform and monitor the fire protection work, and participate in the municipal fire inspection. The Fire safety supervisor shall annually plan for and participate in evacuation exercises within his / her activities.

In case of evacuation, the Fire Safety supervisor will be wearing a yellow vest and is the person to whom the evacuation leaders (in orange vest) report.

8.4 Fire/evacuation Alarm

The SciLifeLab CS buildings are divided into separate fire cells, with each floor acting as an individual fire cell. These fire cells are designed to contain a fire for approximately 30 minutes, providing critical time for evacuation and emergency response.

The fire alarm system is designed to activate on a single fire cell/floor at a time, ensuring targeted and efficient alerting. The alarm can be recognized as a high signal alarm with a red flashing light. This alarm automatically notifies the fire brigade and Akademiska Hus.

When a fire/evacuation alarm is activated, individuals in the affected area must evacuate and gather at the designated assembly point. However, those located on other floors/separate fire cells are recommended to remain where they are until the fire alarm is deactivated, provided it is safe to do so. It is safe to stay in a fire cell where no smoke or heat has been detected and where no fire alarm has been activated.

IN CASE OF A FIRE ALARM

Quickly look around for obvious signs of fire (visible flames, smoke, the smell of smoke, or unusual heat). If you discover a fire, attempt to extinguish it only if you judge it safe to do so; Otherwise, proceed to evacuate. Consider whether you can assume the role of evacuation leader.

IN CASE OF A FIRE

1. **Activate the fire evacuation alarm (if not active yet):**

- Locate the red fire alarm button near all floor entrances and press the button to activate the alarm on your floorNever be scared of using the fire evacuation alarm! It is better to use it too often, rather than not to use it in case of an emergency.

2. **Fight the fire or evacuate:** the decision of whether to stay and fight a fire, or to evacuate, must always be made according to the *type and size of the fire, its location, and the circumstances of the fire:*

    - A small fire can be extinguished by using a fire extinguisher or other suppression equipment available, avoiding bigger damage and losses.

    - Do not stay and fight a large or hazardous fire such as those affecting hazardous or unknown chemicals!

OTHER SITUATIONS THAT REQUIRE EVACUATION

The fire/evacuation alarm may also be activated in emergencies other than fire. These include larger chemical spills, unusual chemical odors, repeated incidents of nausea, or any other situation that could pose a health risk to people on the floor.

Firefighters are trained to handle a wide range of emergencies, including chemical spills. Make sure to inform them about the nature of the evacuation when they arrive and by calling 112.

EVACUATION PROCEDURE

1. Evacuate through the nearest smoke-free escape route. Please be aware that in the Alfa and Gamma buildings, emergency escape doors are located at the bottom of each staircase on the ground level, and these should be used as the primary emergency exits if you are in those buildings.
2. Inform people that you pass to evacuate and remember those who are working in closed areas. Do not use the elevators.
3. On your way out, if it can be done safely, turn off electrical equipment and move any explosive or flammable materials away from possible contact with hot surfaces or other sources of ignition. Close all the doors behind you.
4. Remember to use the emergency handles on your way out. These unlock the doors so that you can get back in without having to use your key card, this might be necessary for example if you need to help your colleagues if they are injured or in panic.

5. If evacuating through the speedgate, to set the speedgate in open position, press the black button in the green box on the left side of the speedgate. Gates will open and an alarm will sound until reset of the speedgate. Site support, reception or security guards will be able to reset.

6. The assembly point is **Fogdevreten 2** (behind Gamma building). Meet up with your evacuation leader and stand close to your group. Keep out of the road and be careful if crossing the street.
7. Attend to injured persons and call the emergency services at 112 if necessary (the fire brigade will show up automatically). Call/email **KTH emergency number 08-790 77 00** and Site Support (support@scilifelab.se) and inform about the evacuation.
8. Do not re-enter the building until the evacuation leader tells you it is safe to do so.
9. Information from SciLifeLab Site Support might be sent to your phone or to your email, keep yourself updated by checking these forums regularly.

8.5 Ventilation Pressure Alarm (Warning)

A blue flashing light with **no siren** indicates that the main ventilation is shut down in the building, that the ventilation has low pressure, or that a fire alarm has gone off somewhere in the building.

IN CASE OF A VENTILATION PRESSURE ALARM;

- The ventilation pressure alarm is **not an evacuation alarm and therefore not considered urgent**. **This alarm is only a warning.** It is safe to stay in the building.

- In case of alarm, do not use or rely on the ventilated stations such as LAF benches, ventilated hoods or ventilated benches until the alarm is reset. Be careful if handling cryogenic liquids, dry ice or gas in enclosed areas.

- Akademiska Hus is automatically alerted when the ventilation pressure alarm has been triggered in the building.

WHAT TO DO IN CASE OF VENTILATION PRESSURE ALARM;

1. Call Akademiska Hus, or contact the Site Support *(see section 9.1 Emergency Contacts in Case of Fire, Evacuation and Accidents).*

2. Do not use ventilated installations or equipment until the alarm is shut off. The alarm will turn off automatically when the ventilation pressure is back to normal.

8.6 Low Oxygen Level Alarm

The localized low oxygen level alarm (Sw: Låg Syrenivå) is independent of emergency services and serves as an alert mechanism within specific areas. When activated, it emits a **high tone signal** and/or triggers a **blue flashing light** outside of designated spaces such as rooms equipped with liquid nitrogen or gas supply systems. These signals indicate that the oxygen concentration within the room has fallen below safe levels. It's important to note that this alarm does not necessitate a building evacuation; the building remains safe for occupancy and work.

IN CASE OF LOW OXYGEN LEVEL ALARM:

1. **Warn your colleagues and exit the room immediately.**
2. **Close the door.** The room's ventilation system is designed to manage potential leaks effectively when the door is properly shut. It's safe to remain in the building.
3. Put the sign “CAUTION. Do not enter” on the door.
4. While keeping the door to the room closed, check the alarm monitor. Normally, multiple oxygen sensors are installed in the room, and the detected oxygen levels are displayed on the monitor.
5. The alarm will cease automatically once oxygen levels return to normal. You can also silence the alarm from the alarm monitor. Note that this action will not deactivate the actual alarm or the flashing blue light.
6. Contact Site Support or the lab safety coordinator and inform about the alarm.
7. **Do not re-enter the room until the alarm has ceased, and the oxygen level has returned to normal (21%).**

RESCUE EQUIPMENT

In the event of a confirmed or suspected serious gas/nitrogen leakage, the affected area must not be entered without adequate rescue equipment encompassing breathing apparatus. A breathing apparatus is stored in a green box located at the Gamma 2 corridor. This equipment should only be utilized by individuals who have received proper training.

8.7 BSL2 Laboratory emergency assist alarm

All BSL2 laboratories at SciLifeLab are equipped with an emergency assist alarm. The alarm is activated from inside the lab to request assistance. A beeping sound sounds locally around the entrance to the lab. If alarming, put in the relevant Personal Protective Equipment and rescue the person with applicable measures e.g., call 112, or follow the specific emergency routine for the BSL2 lab involved. The alarm can be reset by pressing the white reset button by the door inside the lab.

8.8 Emergency Equipment

During your first week at SciLifeLab you should be introduced to all emergency equipment and to the evacuation routes specific to your floor. If you have any questions regarding the emergency equipment, please contact your Safety representative. You should also attend fire safety- and HLR courses within your first year at SciLifeLab, these courses will be arranged by, and announced via both your home university and by SciLifeLab locally.

All emergency equipment and the emergency routes are controlled 4 times per year by the SBA inspectors.

Fire Extinguishers, Sprinklers and Fire Blankets

Fire extinguishers are located by all floor entrances, in the middle of the laboratory corridors, and in the middle of the office areas. The water ("vatten") extinguisher is the primary fire extinguishing device utilized at SciLifeLab CS. It is environmentally friendly, containing no harmful chemicals. This extinguisher releases water mist, which enhances cooling effects and minimizes damage, rendering it effective for extinguishing fires involving fibrous materials like cardboard, paper and wood, as well as grease fires. It is safe to use on electrical devices up to 1000V. Labs at SciLifeLab that handle heavy and flammable chemicals also have carbon dioxide ("koldioxid") and powder ("pulver") extinguishers, which are more suitable for fire in these types of materials.

Additionally, SciLifeLab CS premises are equipped with a sprinkler firefighting system. It consists of a network of pipes containing sprinkler heads that release water when heat from a fire is detected, effectively controlling or extinguishing the flames. It is crucial to maintain a distance of at least 0.5 meter from sprinkler heads to ensure their correct functionality. That means it is not allowed to store boxes or place any other material closer than 0.5 m around the sprinkler head.

Fire blankets are suitable for putting out smaller fires on e.g. lab benches, and they can also be used if a colleague's clothes catch fire. Please remember that when a person is on fire – ask the person to STOP; DROP AND ROLL, and always extinguish the fire starting from the head down! This to avoid facial burns.

If you need any additional types of fire extinguishers on your floor, or if you are unsure about what types of extinguishers you need, please contact Site Support.

Emergency Eye Shower and Emergency Shower

Emergency shower stations are located in the main laboratory corridors and are equipped with an eye shower and an emergency shower. To activate the showers, pull the handles down. Some stations also hold a fire blanket and fire extinguisher. Separate eye rinsing bottles can also be found in most laboratories and/or in the main laboratory corridors. Emergency showers should be used in case of, for example, chemical splashes or chemical burns on clothes or on the skin, or if your clothes catch fire. *For more information on the emergency routine, see section 9.9.*

First Aid Stations

Band aids and other first aid kit equipment can be found by most lab entrances by the hand wash stations. First aid stations can be used for treating minor traumatic injuries such as cuts, stings and burns. For more traumatic injuries, call emergency services on 112.

If components of the first aid kit are finished, missing or expired and need to be replenished, please send a ticket to Site Support.

Defibrillators

Defibrillation is an electrical shock delivered to the heart designed to terminate a life-threatening arrhythmia or cardiac arrest. The Automated External Defibrillator (AED) is a device capable of automatically detecting a heart rhythm that requires a shock.

Defibrillators are located on Delta by the reception, on Alfa 4 by the front door, and on Gamma 4 by the front door

What to do in case a person has an emergency episode:

1. **Confirm cardiac arrest.** You need to check to ensure that it is cardiac arrest before you use an AED. Check to see if the victim is unable to respond, if the person is breathing, and for the pulse. You can use the ABC method (check Airways, Breathing and Circulation). If you find no pulse or breath, you need to start CPR.

2. **Call 112.** Explain to them where you are and what is going on.

3. **Start CPR.** If you are not there alone, you should start giving CPR while the other person is getting the AED. If you are alone, call 112, then start the CPR.

   Give 30 chest compressions and then 2 rescue breaths for every 30 chest compressions. You should give CPR right away if you don't know how long a person has been unconscious, then you should use the AED.

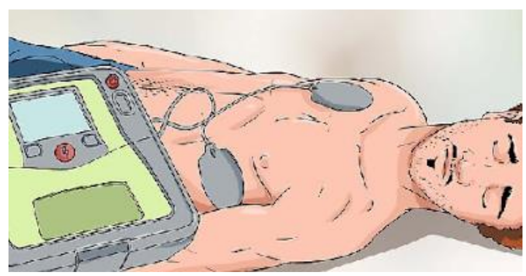
Using the AED:

4. **Make sure the patient is dry.** Water conducts electricity. If the patient is wet or if there is water nearby, the patient can be seriously injured.

5. **Turn on the AED.** It will give you instructions on how to handle the situation. It will likely tell you to attach the cables for the pads into the AED machine. You typically hook them up above the blinking light on the top of the machine. It will also instruct you to prepare the person once the pads are plugged in.

6. **Prepare the chest area.** To use the AED pads, you must remove certain things from the victim. Open or cut through the person's shirt. You should also look for signs of implanted devices, such as a pacemaker. If you see any metal jewelry, a bra or accessory, remove it. The metal will conduct electricity.

7. **Apply the pads.** The electrodes for the AED are adhesive pads. One pad should be placed below the collarbone on the upper right side of the victim's bare chest. The other should be placed below the peck or breast on the left, at the bottom of the person's heart, slightly along the side.



8. **Shock the victim if necessary.** If the AED advises that you need to shock the patient, you need to make sure, once again, that the victim is clear. Once you do, push the *shock* button on the AED. This will send an electric shock through the electrodes to help restart the heart. Stop if the victim can breathe or if the person regains consciousness. If not,

9. **Continue CPR.** Once you have given the victim a shock, you need to continue CPR. You should do it for 2 additional minutes and then let the AED check for a heart rhythm again. Keep this up until emergency services arrive.

8.9 In Case of Chemical Splashes in the Eyes or Chemical Burns

IN CASE OF CHEMICAL SPLASHES IN THE EYES;

1. Call for help from a colleague and wash your eyes immediately with any of the eyewash stations.

2. Consult the chemical Safety Data Sheets for guidance on first aid measures. Wash your eyes for at least 15-30 minutes or longer if recommended by the SDS.

3. Visit the hospital or doctor if needed and recommended by the SDS.

4. Do an incident report together with a Safety representative.

IN CASE OF CHEMICAL BURN;

1. If possible, remove the chemical causing the burn, for example use a brush.

2. Remove all contaminated clothing or jewelry and rinse the area of exposure immediately. Run cool tap water over the burn for at least 10 minutes or use the emergency shower for larger exposures.

3. Call for help from your colleagues and consult the SDS for further guidance.

4. Loosely apply bandages or gauzes over burns.

5. Visit the hospital or doctor if needed.

6. Do an incident report together with a Safety representative.

8.10 In Case of a Chemical Spill

If handled properly, a chemical spill may not cause any significant damage, but if handled improperly, a spill can seriously disrupt your activities, cause bodily harm or property damage.

IN CASE OF A CHEMICAL SPILL;

1. Do a quick evaluation of the situation. Alert your colleagues.

   Is it a large spill (>5Liters), which you cannot clean up, or a spill which you feel uncomfortable handling? Is there anything close to the spill that can make it worse? Any ignition sources? Is there a risk for serious injury, such as an acute respiratory hazard?

If the spill is considered hazardous, evacuate the floor immediately. Use the floor evacuation alarm for "smaller/medium" spills, and the fire evacuation alarm for larger and hazardous spills.

> Close all doors into the lab and put up the "hazardous zone, do not enter" warning signs so that no one enters by mistake. Attend to injured persons and call 112. Contact KTH-Alarm. Contact the SciLifeLab administration. *See 9.1 Emergency contacts.*

If it is a small spill that you feel safe with handling, proceed with the cleanup routine. Chemical clean-up kits are available on every floor in the corridor. You can get advice from an expert from SEKA during office hours (*see 9.1 Emergency contacts)*

Note: *Clean-up the area only if you can do it without any major risk. First decide if you are trained, knowledgeable, and equipped enough to handle the spill! Never clean up a spill alone. Never let the cleaning staff remove chemical spills. Cleaning of chemicals in a fire-damaged area should always be done by a decontamination company.*

CLEAN-UP ROUTINE;

1. Close off the area if necessary. Consult the SDS for guidance.

2. Locate the closest chemical clean-up kit. Use gloves that are suitable for the chemical in question, goggles, boots, and a dustpan or similar. Gas-mask if necessary.

3. Liquid spills should be dried thoroughly with paper or be soaked up using an absorbent material such as vermiculite (Floor Dry), or absorbent pads. Use 4x the amount of vermiculite as the volume of liquid to be absorbed. For powders: use moistened vermiculite, which reduces the risk of aerosols. To use this method, ensure that the chemical being cleaned is non-reactive with H_2O. Use a brush or tongs to remove debris such as glass**.** Do not touch chemicals or glass with your hands!

4. Put the paper/absorbent material and all other collected waste in a yellow laboratory waste bin. Contaminated gloves and glass etc. can be put in the same container. Leave it in a fume hood for potential fumes to evaporate, and then close the lid properly.

5. Put a label on the container with the following information: "Spill of xx" (the name of the chemical) “and absorbed with yy” (the name of the absorbent).

6. The waste should be handled as “chemical waste”.

7. Clean the floor properly with mild detergent to remove any remaining residue. Also wipe off the dustpan before replacing it in the spill kit.

8. Report the event to the nearest Safety representative. Serious incidents should be reported to the Swedish Work Environment Authority within 24h.

IN CASE OF ACCIDENTAL RELEASE OR SPILL INTO THE SINKS OR SEWERS;

Contact the following instances immediately;

- A Safety Representative

- SciLifeLab Administration
- Käppalaverket (Solna): 08-766 67 00 (office hours), 08-766 67 36 (after office hours)
- KI environmental unit: 08-524 801 00 (office hours)
- Solna municipality: 08-746 10 00 (office hours)

If the accidental spill poses a serious hazardous threat to the lab, for example if there is a risk of release of hazardous fumes or the formation of reactions in the sink, you have to handle the situation as a ”large chemical spill” and evacuate the floor immediately.

8.11 In Case of a Biological Spill

In the event of a spill of biological material, it is important to quickly take measures, in order to minimize the harm to staff or persons in the surroundings.

IN CASE OF BIOLOGICAL SPILL;

1. Seal off the area and wait 30 min. This should be done to avoid exposure to potential aerosols and fumes.

2. Read the risk assessment for information on PPE and how to decontaminate.

3. Put on the PPE and decontaminate, if possible. Never let the cleaning staff clean up spill! They do not have the proper training.

4. Most often, the area should be covered with paper towels or equivalent material, after which a disinfectant chemical shall be poured onto the paper towels. It is important that the chemical has a sufficient contact time, i.e., 5-10 minutes. Thereafter, the material shall be handled as contagious. The surface should be disinfected again, and then washed with a non-oxidizing chemical, such as ethanol or detergent. Note that the disinfected chemical shall be proven effective against the biological material in question.

5. Remove all PPE, wash and disinfect your hands.

6. In the event of contamination of a centrifuge or incubator, the lid/incubator door shall instead be closed, in order for any aerosols to settle before it is cleaned in accordance with points 2-5 above.

7. Contact the occupational health service, infection clinic or the biosafety administrator for advise.

8. Contact your closest Safety representative and write an incident report.

9 Hazardous Waste

Below you can find a summary of the most common waste types that we have in the laboratory.

The waste room for hazardous waste at SciLifeLab is located on floor 1, next to the delivery dock. There are several smaller rooms in the main waste room, each designated for a specific type of waste.

Bins, sharps containers, cardboard boxes and bags, absorbent pads, and labels are provided by Site Support and can be picked up from the main waste room.

At SciLifeLab we have the following hazardous waste-types, designated waste bins and waste rooms;

Type of waste	**Designated color of the waste bin**	**Waste room**
Sharp and/or Infectious	Yellow waste bins	Main room
Pharmaceutical and cytostatic	Yellow waste bins	Main room
Chemical	Cardboard Boxes with a black plastic bag or original package	B1161, B1160 or B1121*
Chemical Solvents	Suitable Plastic/Glass containers or original package	B1161, B1160 or B1121*
Biological	Black waste bins	-20°C Freezer for Biological waste in the main room
Radioactive	Green waste bins	Room B1140 for radioactive waste. Code locked door. *

* <u>Room B1161</u> for chemical waste classified as *corrosive –acids or bases-* or *Acute toxicity* and *serious health hazard.* Chemical waste must be placed in the designated safety cabinets.

Room B1160 for chemical waste classified as *oxidising* or *flammable*. Chemical waste must be placed in the designated safety cabinets.

Room B1121 for chemical waste classified as *Corrosive (no acids/bases)*, *Health hazard (exclamation mark)* or *Hazardous to the environment.* Cardboard boxes with consumables contaminated with chemicals are also placed in this room.

Room B1140 for radioactive waste.

The person that produces hazardous waste is responsible for its proper packaging, labelling and disposal.

As a general rule, all gloves used in the lab should be disposed of as hazardous waste.

The following apply for waste bins/boxes/containers:

- The bin must be labeled with a fully completed label. It is advisable to affix the label to the bin at the time it is put into use.
- If a bin is to be filled with liquid waste, the liquids should first be collected in an appropriate container with a tight lid, and an absorbent pad must be placed at the bottom.
- The bin must not weigh more than 12 kg. Do not overfill the bin.
- The lid must be sealed carefully.
- The outside of the bin must be clean. The personnel handling the waste do not wear gloves or other safety protective equipment.
- The bin must be placed in its designated room, in the waste room floor 1.

9.1 Sharps and Infectious waste – "Stickande/Skärande smittförande avfall"

Infectious waste includes human blood and blood products, microorganisms, cell cultures and materials that have come into contact with these waste types, for example gloves, inoculation loops, pipette tips, pipettes, tubes, paper towels.

This also applies when the waste is in addition contaminated with small amounts of pharmaceuticals or chemicals.

Sharp waste includes all sharp objects such as needles, scalpels, lancets, suture needles and microscope slides. This applies even if the material is not suspected to be infectious.

- Sharp and/or Infectious solid waste must be collected and placed in a **yellow box**.
- Sharps can be first collected in smaller sharp safe container
- Infectious liquid waste must first be collected in **sealable bottles or containers** and then placed in a **yellow box** with an absorbent pad at the bottom.

- Label the box with the label “SKÄRANDE/STICKANDE SMITTFÖRANDE AVFALL” and fill in all the fields.

How to fill in this label: *Skärande/Stickande smittförande avfall* = sharp and infectious waste. *Sjukhus/motsv* = SciLifeLab-University. *Avd/motsv* = research group/facility. *Tel* = your telephone for quick contact in case of an incident. *Namn* = your name (the person handling/producing the waste, who knows the content). *Datum* = date of disposal.

Note: “Sharps/infectious waste” labels must NOT be used for inactivated/sterilised waste.

Note: Antibiotics/cytostatics/chemicals are often used in the same experiment as infectious agents. This mixed waste can always be handled as Sharps/infectious waste; however, even though it has been inactivated/sterilised, due to its pharmaceutical/chemical content it may not be suitable for treatment as combustible waste. In these cases, the inactivated/sterilised waste is handled as either “Pharmaceuticals, including cytostatic waste” or as “Chemical waste”.

Exceptions

- **Solid** waste that has come into contact with well-characterised cell cultures that are neither genetically modified nor infectious. Liquid waste is NOT exempt from the regulation.

- Inactivated/sterilised waste, but only if it has been inactivated/sterilised using an approved method.

Approved methods of inactivation/sterilisation

i. Autoclaving. This method kills all infectious waste including spores and can always be used as a sterilization method.

ii. Heat/chemicals/other treatment. This method is only permitted if it has been demonstrated to be effective against the infectious agent in question.

9.2 Pharmaceutical and cytostatic waste – “Cytostatika och läkemedelsförorenat avfall”

Pharmaceuticals and cytostatics can pose environmental risks if incorrectly handled. These substances are often slow to degrade and can be biologically active for a long time after use. They can also be highly toxic, induce mutations or cause hormonal disruptions.

Pharmaceuticals, including cytostatic waste includes pharmaceuticals such as vaccines, antibiotics, cytostatics, narcotic drugs, hormone preparations and objects that come into contact with these materials, for example packaging that contained antibiotics, cytostatics or other pharmaceuticals.

- Waste that has been contaminated with cytostatic and other pharmaceuticals must first be placed in a sealed inner packaging (sealed plastic bag or similar) before being placed in a **yellow box** with other pharmaceutical waste.

- Liquid pharmaceutical waste must be collected first in leak-proof, sealable bottles/containers and then placed in a **yellow box.**

- The label for “Pharmaceuticals, including cytostatic” consists of two labels; the hazard symbol for “Acute toxicity” together with the hazard symbol for “Hazardous to the environment”. Label the box with “CYTOSTATIKA OCH LÄKEMDELSFÖRORENAT AVFALL” and fill in all fields.

How to fill in this label: *Cytostatika-läkemedelsavfall samt läkemedelsförorenat avfall* = cytostatic-pharmaceutical waste and pharmaceutical contaminated waste. *Sjukhus/motsv* = SciLifeLab-University. *Avd/motsv* = research group/facility. *Tel*= your telephone for quick contact in case of an incident. *Namn* = your name (the person handling/producing the waste, who knows the content). *Datum* = date of disposal.

Narcotic drug waste must be de-identified before being placed with other pharmaceutical waste. Disposal must be documented in the hospital drug consumption register. Documentation must be witnessed by two authorised persons.

Needles and other **sharp waste** must be handled as sharp and infectious waste even if contaminated with pharmaceuticals

Some drugs can be subjected to **additional regulations**, for more information contact the Lab Safety Coordinator.

For "**antibiotics used in cell cultures**" see the "Chemical waste" section.

9.3 Biological Waste – Biologiskt avfall

For ethical reasons, biological waste must be handled separate from other waste. Biological waste includes larger samples of body parts, tissues, teeth and organs from people and animals as well as anatomical preparations and similar - but only if they do not contain sharp objects. Biological waste must always be stored frozen.

- **Biological waste** must be placed in **black boxes**. The material must be placed in plastic bags before placing it in the box. Absorbents should be placed in the bottom.
- The **black box** must be labelled with a fully completed "BIOLOGISKT AVFALL" label. Regardless of whether the animal has been exposed to GMM, microorganisms at Level 2 or a pharmaceutical, only the label "BIOLOGISKT AVFALL" is to be used.
- Leave the boxes in the waste room floor 1, in the freezer for biological waste.

How to fill in this label: Biologiskt avfall = Biological waste. Sjukhus/motsv = SciLifeLab-University. Avd/motsv = research group/facility. Tel = your telephone for quick contact in case of an incident. Namn = your name (the person handling/producing the waste, who knows the content). Datum = date of disposal.

Note: Sharp waste (e.g tissue slides) and all consumables that come into contact with biological material, are handled as "Sharps/infectious waste".

9.4 Chemical Waste - Kemikalieavfall

Chemical waste includes:

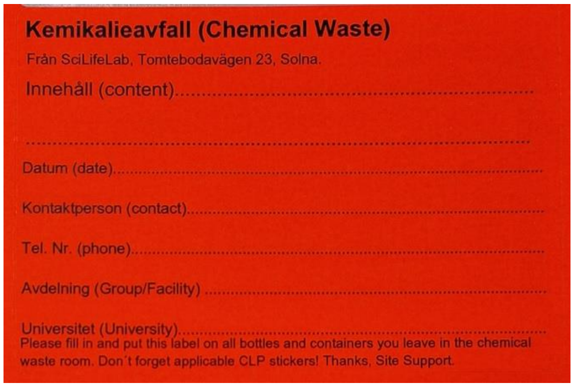
- Chemicals marked with hazard symbols, including reagent solutions, solvents, oils, paints, adhesives, disinfectants, etc.
- Unidentified chemicals (must be denoted with “Unknown chemical” under “Contents”).
- Materials such as gloves, paper towels and test tubes, that are contaminated with chemicals labelled with the hazard symbol for “Acute toxicity”, “Serious health hazard” and/or “Hazardous to the environment”.
- Bottles and cans labelled with the symbol for “Acute toxicity”, “Serious health hazard” and/or “Hazardous to the environment”. This also applies to empty containers.
- Photochemical waste; developers, fixing solutions and films.
- Certain objects containing environmentally hazardous metals such as mercury thermometers and lead aprons from radiography.

Note: Diluting chemicals to negligible concentrations or removing chemicals via evaporation in a fume cupboard is not permitted.

Note: Bottles containing non-hazardous solid chemicals (i.e., without hazard pictograms or H-statements), such as certain culture media powders or NaCl , are not classified as hazardous waste. These can be disposed of according to their material type – for example, as combustible waste, plastic, glass, or other appropriate waste categories.

Chemical waste can be disposed of in its original packaging or other suitable container (glass bottle/plastic bottle/etc.) preferably of the same type of material that the chemical was originally stored in that is not affected by the contents. For liquid chemical waste, use the original packaging or an appropriate container, e. g. UN-approved container for transport of dangerous goods. Never mix chemicals of different types i.e. acids should never be mixed with bases etc. Some liquid chemicals can be mixed, but verify this before you combine them by reading the Safety Data Sheets. The container must be properly sealed and the lid must be intact.

Use a completed fill out “Kemikalieavfall” label directly on the container. Mark the container with the hazard pictograms. If the container has the original markings indicating the correct content and pictograms, they do not need to be indicated again.



Kemikalieavfall (Chemical Waste)

Från SciLifeLab, Tomtebodavägen 23, Solna.

Innehåll (content)..

....

Datum (date)..

Kontaktperson (contact)...

Tel. Nr. (phone)...

Avdelning (Group/Facility) ....

Universitet (University)...

Please fill in and put this label on all bottles and containers you leave in the chemical waste room. Don´t forget applicable CLP stickers! Thanks, Site Support.

How to fill in this label: *Kemikalieavfall (*chemical waste). *Datum*(date) = date of disposal. *Innehåll (*content) = short description of the chemical waste. *Kontakpersonperson* (contact)= your name, the person handling/producing the waste, who knows the content. *Tel. Nr*. (phone) = your telephone number for quick contact in case of an incident.

If you are getting rid of a lot of solid chemicals, you may put the containers in a **cardboard box**, and label with the label for chemical waste. Please then just be sure to separate between different types of chemicals.

Halogenated chemical waste

To ensure proper environmental reporting and cost-effective disposal, collect and label all halogenated waste (organic chemicals with F, Cl, Br, or I) separately as "Halogenated Solvent Waste"

Antibiotics used in cell cultures

Several common antibiotics are classified as non-biologically degradable which means that release to the drains entails a risk to the environment. Autoclaving does not result in complete degradation of the majority of antibiotics, and the degradation products that are formed can in turn have negative impacts on the environment. **Discharge of antibiotic waste to the drains is therefore not permitted.**

Provided that there is no contamination with live cells, antibiotic waste shall be handled as **Chemical waste**. Choose a suitable container and label it “Antibiotic-containing solution”. The container does not have to be UN-labelled. No hazard pictograms are required unless the waste contains additional chemicals that require such labelling. Liquid antibiotic waste will be processed using a method that evaporates and purifies the liquid, and any solid residue is sent for incineration.

Cell culture medium that contains antibiotics where the live material has not been killed or which might contain infectious material must be handled as “Sharps/ infectious waste”.

Contaminated Consumables and Sharps

Material, sharps and consumables contaminated with chemicals labelled with the symbol for “Acute toxicity”, “Serious health hazard” and/or “Hazardous to the environment” must be put in a cardboard waste box with an inner thick black plastic waste bag. **All sharps should in all cases first go into a sharps container**, and then put into the cardboard box. These cardboard boxes are mainly used in our chemical laboratories for the waste from the fume hoods, and these are kept ventilated. If the box is not ventilated while in use, first place the contaminated material in a sealed plastic box in order to avoid exposure to fumes or contaminants. When full, use the label for chemical waste.

Note: yellow bins should not be marked with the label for chemical waste:

Exception: Small quantities of material (not packaging) that have been contaminated with chemicals labelled with the symbol for “Highly toxic”, “Carcinogenic/Mutagenic” and/or “Environmentally toxic” can be dealt with as “Sharps/infectious waste”. **Only if there is no risk of exposure!**

Empty Bottles / Cans

Bottles and cans labelled with the hazard symbols “Acute toxicity”, “Serious health hazard” and/or “Hazardous to the environment” must always be handled as “Chemical waste” also when empty. Normal recycling centers will not accept packaging with such labels.

Clean packaging that does not have the above hazard symbols shall be handled as normal waste, i.e. sorted as glass, hard plastics, etc. – but only if it does not contain residues, dust or droplets! Sealed packaging that has contained flammable goods must be opened and rinsed carefully with water to prevent the contents giving off flammable vapor after the packaging is placed in the source sorting bins. The packaging shall be discarded without a lid to further minimize the risk of fire.

In cases of doubt, the packaging shall be handled as chemical waste.

Empty Gas cartriges

Empty gas cartridges must be disposed of as hazardous waste. A maximum total of 1 L (container volume) may be placed in the fire safety cabinet for flammable waste to be collected by the hazardous waste disposal company.

Special Handling

- Containers that contain chemicals for which a permit is required (A/B classified) must be labelled with the chemical name and that it is a regulated chemical.
- For highly flammable substances such as butyl lithium etc., seek guidance from the SciLifeLab Lab Safety Coordinator.

Storage of chemical waste

Storage of chemical waste is governed by the same regulations as chemicals in general which means that:

- Ventilation must be sufficient to avoid accumulation of vapour.
- Liquid chemical waste must not pass to the sewers via floor drains or similar.
- Containers with liquid chemical waste must be stored encapsulated or in trays that contain the entire volume of the largest container + 10% of the remaining total volume of the chemicals that are stored in the tray.
- Substances that can react with each other must not be placed in the same drip tray; for example:

  * acids must not be stored together with bases,

  * ammonia must not be stored together with hydroxides and hypochlorites,

* oxidizing substances must not be stored together with oxidisable substances.

- Chemical waste must be stored so that it is inaccessible by unauthorised persons.

- If the room for chemical waste is situated separately, emergency routines should be posted and decontamination systems and appropriate protective equipment must be available in the premises.

Discharge to the Drains

Disposing chemicals subject to “compulsory labelling regulations” (= all chemicals that have hazard statements and hazard pictograms) down the drain / sink is generally forbidden;

General rule:

- **No Organic or Volatile Solvents** are allowed to be poured down the drain! (e.g Xylene, ethanol, methanol, Acetone)

- **No environmentally hazardous chemicals** are allowed to be poured down the drain! (E.g. Sodium Azide, cobalt II chloride, 2 mercaptoethanol, phenol/chloroform)

- **No Toxic or Hazardous chemicals** are allowed to be poured down the drain! (e.g. Paraformadelhyde, Acrylamide, potassium bromate)

The drain water from Campus Solna goes directly to Käppalaverket. The drain water can only be processed under condition that it does not vary significantly from normal household waste water.

- All **pure chemicals** and **preparations whose chemical content distinguishes them from normal household drain water** must be collected in containers and sent as chemical waste for destruction. It is not permitted to pour live microorganisms/cell cultures into the drains!

- In the event of **major spills or discharge of chemicals to the drains**, the sewage treatment plants and municipality must be contacted promptly. *See section 9.10 "in case of a chemical spill" for more information.*

- Only solutions that are known, without any doubt, to be **harmless** to the treatment plants’ various processes as well as to the Baltic Sea and to its organisms, both in the short and long term, can be discharged to the drains.

- **Nutrient solutions** that only contain vitamins, electrolytes, amino acids, peptides, proteins, carbohydrates and lipids are not considered to result in any significant effects on the environment. Residues from such nutrient solutions can be poured into the drains.

Note that a product in a concentrated form may need to be handled as hazardous waste even though the product in solution can be rinsed to the drain during use. For example, unused dishwasher tablets/powder must be handled as hazardous waste while the discharge from the dishwasher passes into the drains. Disinfectants can also contain substances that are harmful for aquatic organisms, for example Gigasept, Chemgene, and Virkon, and discharge of these solutions to the drain must be only limited.

If a chemical product with mandatory labelling is to be discharged into the drain, all three of the points below must be met:

1 - **Only small amounts (<10 ml),** for example experiment residues or solvents that are difficult to collect when dishwashing glassware.

2 - **The pH must not be lower than 5 and no higher than 11.5** - highly acidic or basic solutions can damage the pipe systems. Substances with the Corrosive hazard symbol can be poured into the drains if they have no other hazardous properties and if the pH has been adjusted.

3 - **The chemical is neither an environmental hazard nor harmful i.e. it cannot be marked with hazard codes H400-H413**, such as heavy metals and certain organic substances that are difficult to degrade, toxic, bio accumulative (stored in living organisms) or affect nitrogen separation in the treatment plant.

4 - The chemical may **not be cancerogenic, mutagenic or toxic to reproduction** i.e. it cannot be marked with hazard statements H340, H341, H350, H351, H360 or H361.

If uncertain, apply the safety principle and handle it as chemical waste.

9.5 Radioactive Waste

Requirements related to handling radioactive waste can be found in SSMFS 2018:1.

Radioactive waste is subjected to additional regulations.

- Radioactive waste must be dealt with separately and must not be disposed of with normal waste.
- The green waste bins should be used for all radioactive waste.

- There is a special room in the basement waste room designated for radioactive waste, this door has a code-lock.

Contact the SciLifeLab Lab Safety Coordinator for information on radioactive waste handling.

10 Recycling

Paper, cardboard, plastic, metal, glass, combustible, lamps, electronic waste, toners and others are disposed of at the environmental stations. Fully equipped environmental stations are located in A2592, A3200, A4200 and G4590. Smaller environmental stations, with designated containers for the most common waste, are located in A1790, A5200, A6330, G3770, G5360, G6700. Only nonhazardous and non-contaminated waste is allowed in these rooms. All hazardous waste should be treated as laboratory waste. The SciLifeLab's main sorting facility is in the main waste room at floor 1 and is equipped with containers for all fractions.

The service company Coor empties the environmental stations at regular intervals or when needed. Issues or requests for new containers are addressed to Site Support.

More details about the different sorting fractions can be found on KI's website: Waste Management.

APPENDIX 1. Introduction Checklist

SciLifeLab – New Personnel Introduction Checklist

Name of new member: ____

Supervisor: ____

Group leader: ____

Date for introduction: ____

All new employees and students should receive a proper introduction to their new workplace. The group leader appoints a supervisor (either themselves or another suitable person) for the introduction. Together, they are responsible for making the onboarding welcoming, informative and useful.
This checklist is designed as a guide for introducing new colleagues at SciLifeLab. The checklist should be kept with the new employee.

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1. Administrative Matters (arranged via Site Support)

- ☐ Access card (group leader applies via SciLifeLab Staff database, new employee collects it in the Service Desk)
- ☐ E-mail account (group leader applies via the SciLifeLab Staff Database)
- ☐ Keys (office/locker, collected from Servicedesk)
- ☐ Read, sign, and submit the SciLifeLab policy document to Service desk (required before access card and accounts are granted)

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2. Human Resources

- ☐ Provide contact information to their home university HR (for issues regarding employment contract, salary, vacation, wellness etc.)

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3. Technical Matters (home university responsibility)

- [ ] Telephone and phone number
- [ ] Computer

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4. Organizational Introduction

- [ ] Send a welcome e-mail (with photo) to "your floor"@scilifelab.se
- [ ] Update staff database profile with photo and information (staff.scilifelab.se/mypage)
- [ ] Sign up for floor and utility mailing lists (via staff database) and Slack

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5. Intranet (intranet.scilifelab.se)

- [ ] Tour and demonstrate key features:

  - Support at Campus Solna (right menu)
  - Report incident (right menu)
  - Personnel search and contact details
  - Room & instrument booking
  - Floor maps
  - Responsibility areas and appointed persons
  - IT

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6. Safety

New personnel must be informed of all risks specific to their group's activities before starting any work.

Training and information

- [ ] Provide the Workplace handbook
- [ ] Sign up for the first available Safety Introduction
- [ ] Sign up for the first available Fire Safety training
- [ ] Show first aid stations & CPR equipment
- [ ] Review fire evacuation plan & fire extinguishing equipment
- [ ] Show evacuation routes, emergency exits and assembly point

Waste Management

- ☐ Waste disposal instructions and sorting guides (on intranet)
- ☐ Environmental station on your floor
- ☐ Main recycling room (Basement)

Chemicals

- ☐ KLARA chemical database & how to retrieve SDSs
- ☐ Risk assessments – location and preparation

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7. Tour

- ☐ Reception and Service Desk
- ☐ Conference area (incl. Air&Fire)
- ☐ Lunchrooms (G2, G7)
- ☐ Changing rooms (G2, A1)
- ☐ Resting rooms (Basement, G6, G7)
- ☐ Main recycling room (Basement)

Date & Signature of New Member	**Date & Signature of Group Leader**
______________________________	______________________________